Domain Analyses: Growth Trajectories and Open Problems I

This supplementary section provides detailed characterizations of each of the eight tracked Active Inference domains, organized under three tiers: A (Core Theory), B (Tools & Translation), and C (Application Domains).

Domain Analyses: Growth Trajectories and Open Problems II

Domain A: Core Theory

A1 — Quantitative & Formal Theory ($n = 64$, 7.8%)

The A1 domain develops the mathematical foundations underpinning the Free Energy Principle: information geometry, category-theoretic formulations of Markov blankets, path integral formulations of free energy minimization, and gauge-theoretic perspectives on self-organization. A central debate concerns the ontological status of Markov blankets—whether they correspond to real physical boundaries or are merely useful statistical constructs [bruineberg2022emperor]. Bruineberg et al. draw a critical distinction between *Pearl blankets* (instrumental, epistemic tools for conditional independence in Bayesian networks) and *Friston blankets* (ontologically laden physical boundaries between agent and environment), arguing that the scientific credibility of the former should not be extended uncritically to the latter. Friston